BIO505P – ESSENTIALS OF GENETICS

Practical Exam Notes

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SHORT QUESTIONS

1. Explain DNA extraction and its function.

DNA extraction is the process of isolating and purifying DNA from cells or tissues. It involves breaking the cell and nuclear membranes, removing proteins and other impurities, and collecting the DNA in pure form. Common steps include cell lysis, removal of contaminants, DNA precipitation using alcohol, and purification.

Functions of DNA Extraction

- Used in genetic and molecular biology research
- Helps in diagnosis of genetic and infectious diseases
- Important for forensic investigations and DNA fingerprinting
- Used in biotechnology and genetic engineering
- Helps in paternity testing and ancestry studies
- Assists in studying heredity, evolution, and species identification

2. What material is required for cDNA synthesis?

Materials Required for cDNA Synthesis

- Template RNA (usually mRNA)
- Reverse transcriptase enzyme
- Primers (oligo-dT, random primers, or gene-specific primers)
- dNTPs (DNA nucleotides)
- Reaction buffer
- RNase inhibitor
- Nuclease-free water
- Thermal cycler or water bath for incubation

3. Write a note on colony PCR.

Colony PCR

Colony PCR is a rapid molecular technique used to identify bacterial colonies containing the desired DNA insert. Instead of isolating plasmid DNA, a small part of a bacterial colony is directly used as the template for PCR.

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Procedure

- A small bacterial colony is picked with a sterile toothpick or loop.
- The colony is mixed in PCR reaction mixture containing primers, DNA polymerase, dNTPs, and buffer.
- PCR amplification is performed in a thermal cycler.
- The PCR product is analyzed by gel electrophoresis.

Applications and Importance

- Confirms presence of recombinant DNA or cloned gene
- Saves time by avoiding plasmid extraction
- Widely used in cloning and genetic engineering experiments
- Helps select positive colonies for further study

4. What is the most common calculation used for purity of DNA in spectrophotometer?

The most common calculation used for determining DNA purity in a spectrophotometer is the A260/A280 ratio.

$$\frac{A260}{A280}$$

- A260 measures nucleic acids (DNA/RNA)
- A280 measures protein contamination

Interpretation

- Ratio around 1.8 = pure DNA
- Ratio lower than 1.8 = protein contamination
- Ratio higher than 2.0 = possible RNA contamination

5. What is the purpose of Phase Separation in extraction of Protein?

Phase separation in protein extraction is used to separate proteins from other cellular components such as DNA, RNA, lipids, and debris. It helps obtain purified proteins for analysis.

Purpose of Phase Separation

- Separates proteins into a specific layer or phase
- Removes contaminants and impurities
- Improves protein purity and yield
- Facilitates further biochemical and molecular studies

6. How is the DNA pellet diluted in isolation of DNA by organic method?

After DNA precipitation and washing in the organic method, the DNA pellet is air-dried to remove alcohol. It is then dissolved in TE buffer or sterile distilled water by gentle mixing or incubation. TE buffer protects DNA from degradation and maintains its stability for storage and further analysis.

7. Write step by step preparation of 5M sodium chloride solution.

Preparation of 5M Sodium Chloride (NaCl) Solution

1. Weigh 292.2 g of sodium chloride (NaCl).
2. Add the NaCl to about 800 mL of distilled water in a beaker.
3. Stir until the salt completely dissolves.
4. Transfer the solution to a 1 L volumetric flask.
5. Add distilled water to make the final volume 1000 mL (1 L).

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8. Write a note on Hot start PCR.

Hot Start PCR

Hot Start PCR is a specialized form of PCR in which the DNA polymerase enzyme remains inactive at room temperature and becomes active only after heating during the initial denaturation step. This prevents unwanted amplification and primer-dimer formation before the PCR cycling starts.

In this method, modified polymerases, antibodies, or chemical inhibitors are used to block enzyme activity at low temperatures.

Advantages

- Increases specificity and accuracy of PCR
- Reduces non-specific amplification
- Prevents primer-dimer formation
- Produces better DNA yield and cleaner results

9. Write step by step process for isolation of RNA.

Step by Step Process for Isolation of RNA

- Collect fresh cells or tissue sample under sterile conditions.
- Add lysis buffer or TRIzol reagent to break the cells and release RNA.
- Homogenize or mix the sample properly to form a uniform solution.
- Add chloroform and shake gently.
- Centrifuge the mixture to separate it into layers.
- Carefully transfer the upper aqueous layer containing RNA into a new tube.
- Add isopropanol to precipitate the RNA.
- Centrifuge again to obtain the RNA pellet.
- Wash the pellet with 70% ethanol to remove impurities.
- Air-dry the pellet for a few minutes.
- Dissolve the RNA pellet in RNase-free water or buffer.
- Store purified RNA at −20°C or −80°C for further use.

10. What material and reagents are required during Karyotyping?

Materials and Reagents Required for Karyotyping

- Blood sample or other dividing cells
- Culture medium
- Incubator
- Colchicine or colcemid
- Hypotonic solution (usually KCl)
- Fixative solution (methanol and acetic acid)
- Microscope slides
- Giemsa stain
- Microscope
- Centrifuge
- Pipettes and test tubes

11. What is the purpose of centrifugation?

Purpose of Centrifugation

Centrifugation is a laboratory technique used to separate components of a mixture based on differences in density by spinning samples at high speed.

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Uses

- Separates solid particles (pellet) from liquid (supernatant)
- Helps in isolation of DNA, RNA, proteins, and cell organelles
- Removes cell debris and impurities during extraction
- Concentrates biological samples for further analysis
- Widely used in blood separation (plasma and cells) and molecular biology experiments

12. What is the purpose of electrophoresis?

Purpose of Electrophoresis

Electrophoresis is a technique used to separate and analyze charged biomolecules such as DNA, RNA, and proteins based on their size and electrical charge by applying an electric field through a gel medium (usually agarose or polyacrylamide).

Uses

- Separates DNA, RNA, or protein fragments according to size (smaller move faster)
- Helps determine the size and purity of DNA fragments
- Used to analyze PCR products and restriction enzyme digestion results
- Important in DNA fingerprinting and forensic identification
- Used in genetic research, disease diagnosis, and biotechnology studies
- Helps compare samples in molecular biology experiments

13. Define southern blotting process.

Southern Blotting (Definition)

Southern blotting is a molecular biology technique used to detect a specific DNA sequence within a complex DNA mixture by using a labeled complementary DNA probe.

Process

- DNA is extracted and cut into fragments using restriction enzymes.
- Fragments are separated by agarose gel electrophoresis.
- DNA in the gel is denatured into single strands using alkali treatment.
- DNA is transferred (blotted) onto a nylon or nitrocellulose membrane.
- The membrane is baked or UV-treated to fix DNA.
- A labeled DNA probe is added and allowed to hybridize with the target sequence.
- Excess probe is washed away.
- Detection is done using autoradiography or fluorescence.

14. Name three free primer designing tools.

Free Primer Designing Tools

- Primer3
- NCBI Primer-BLAST
- OligoCalc

15. Write function of Proteinase K during DNA extraction.

Function of Proteinase K in DNA Extraction

Proteinase K is an enzyme used to digest and remove proteins during DNA extraction.

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Functions

- Breaks down proteins (especially histones) bound to DNA
- Inactivates DNases that can degrade DNA
- Helps release pure, high-quality DNA
- Improves efficiency of DNA extraction process

16. Write principle of Northern Blotting.

Principle of Northern Blotting

Northern blotting is based on the principle of nucleic acid hybridization. In this technique, RNA molecules are first separated according to their size using gel electrophoresis. The separated RNA is then transferred onto a nylon or nitrocellulose membrane where it is immobilized. A labeled complementary DNA or RNA probe is added, which binds specifically to the target RNA sequence through base pairing. The presence of hybridized RNA–probe complexes is then detected using autoradiography or fluorescence. This allows identification, size estimation, and expression analysis of specific RNA molecules.

17. Write steps for DNA precipitation.

Steps for DNA Precipitation

- Collect the DNA-containing aqueous solution after extraction.
- Add salt solution (e.g., sodium acetate or NaCl) to neutralize DNA charge and help aggregation.
- Add cold ethanol (95–100%) or isopropanol slowly to the solution.
- Mix gently by inverting the tube (do not vortex).
- Incubate at -20°C or on ice for 15–30 minutes to enhance precipitation.
- Centrifuge the sample at high speed to form a DNA pellet.
- Carefully discard the supernatant without disturbing the pellet.
- Wash the pellet with 70% ethanol to remove salts and impurities.
- Centrifuge again and remove wash solution.
- Air-dry the pellet to remove residual alcohol.
- Dissolve DNA pellet in TE buffer or nuclease-free water for storage and use.

18. What is repeat? Hint: Primer designing

Repeat (in Primer Designing)

In primer designing, a repeat refers to a DNA sequence in which a short nucleotide pattern is repeated multiple times consecutively or at different locations in the genome.

Types

- Tandem repeats (e.g., ATATAT)
- Interspersed repeats (scattered in genome)

19. Why it is important to use sterilized equipment in the DNA extraction from whole blood?

Importance of Sterilized Equipment in DNA Extraction from Whole Blood

- Prevents contamination from foreign DNA, bacteria, or environmental microbes.
- Avoids introduction of DNases and RNases, which can degrade DNA and reduce yield.
- Ensures high purity and integrity of extracted DNA for accurate results.
- Prevents cross-contamination between different blood samples.
- Maintains reliability, accuracy, and reproducibility of experimental results.
- Essential for sensitive applications like PCR, sequencing, and forensic analysis.
- Ensures valid interpretation in clinical diagnosis and genetic testing.

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20. What are components of cDNA?

Components of cDNA

- Single-stranded cDNA (ss-cDNA) synthesized from mRNA
- Double-stranded cDNA (ds-cDNA) formed after second strand synthesis
- Exonic (coding) sequences only (no introns)
- Reverse transcriptase enzyme (used in formation, not part of final product)
- Primers (oligo-dT/random primers used during synthesis)

21. Why buffer is used in cDNA synthesis?

Why Buffer is Used in cDNA Synthesis

- Maintains a stable pH (usually around 7.5–8.3), which is essential for the activity of reverse transcriptase enzyme.
- Provides proper ionic strength (salts like Mg^{2+}, K^{+}) required for enzyme function and primer binding.
- Creates an optimal reaction environment for efficient conversion of RNA into cDNA.
- Stabilizes the enzyme and prevents loss of activity during the reaction.
- Helps protect RNA from degradation during the synthesis process.
- Improves reaction efficiency, yield, and accuracy of cDNA formation.

22. What role Trizol plays in extraction of RNA?

Role of TRIzol in RNA Extraction

- TRIzol is a strong denaturing reagent used to lyse cells and dissolve cellular components.
- It contains phenol and guanidinium isothiocyanate, which break membranes and inactivate RNases.
- Helps in complete release of RNA, DNA, and proteins from the sample.
- After adding chloroform and centrifugation, it causes phase separation:
    - RNA remains in the upper aqueous phase
    - DNA stays in the interphase
    - Proteins move to the organic phase
- Protects RNA from degradation and ensures high purity and yield.
- Widely used for efficient RNA isolation in molecular biology and gene expression studies.

23. Why we use DNase-free water?

Why DNase-free water is used

- Prevents degradation of DNA by DNase enzymes present in normal water.
- Ensures high purity and integrity of DNA/RNA samples.
- Avoids false or poor results in PCR and other molecular techniques.
- Maintains accuracy and reliability of experimental outcomes.
- Essential for sensitive applications like PCR, sequencing, and cDNA synthesis.

24. Write use of chloroform in Trizol-based extraction.

Use of Chloroform in TRIzol-Based Extraction

- Chloroform is used to separate the mixture into different phases during RNA extraction.
- After adding chloroform and centrifugation, it causes phase separation:
    - RNA remains in the aqueous (upper) layer
    - DNA stays in the interphase
    - Proteins go to the organic (lower) layer
- Helps in clean separation of RNA from DNA and proteins.
- Improves the purity and quality of isolated RNA.

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25. Write uses of Giemsa stain.

Uses of Giemsa Stain

- Used in karyotyping to stain chromosomes and produce distinct G-banding patterns, helping in identification of chromosomal abnormalities.
- Detects blood parasites such as Plasmodium species (malaria) by staining parasites inside red blood cells.
- Used in cytogenetics and cell biology to study cell structure and chromosome morphology.
- Helps in diagnosing bacterial infections like *Helicobacter pylori* in tissue samples.
- Used in bone marrow examination to study different types of blood cells.
- Important in identifying blood disorders such as leukemia and other hematological diseases.

26. What does SDS stands for? What are its uses?

SDS Full Form: Sodium Dodecyl Sulfate (SDS)

Uses of SDS

- SDS is an anionic detergent used to break open cell and nuclear membranes during lysis.
- It solubilizes membrane lipids and proteins, helping release DNA, RNA, and cellular contents.
- It denatures proteins by disrupting hydrogen bonds and hydrophobic interactions.
- In DNA extraction, it helps remove proteins from DNA, improving purity of DNA.
- In SDS-PAGE, it gives proteins a uniform negative charge so they separate based on size only.
- Widely used in molecular biology for cell lysis, protein analysis, and nucleic acid purification.

27. Write composition of 50x TAE buffer.

Composition of 50X TAE Buffer

- ***Tris base*** – 242 g
- ***Glacial acetic acid*** – 57.1 mL
- ***0.5 M EDTA (pH 8.0)*** – 100 mL
- ***Distilled water*** – make final volume up to 1 liter

28. Write advantages and disadvantages of Wester Blotting.

Advantages of Western Blotting

- Highly specific technique because it uses antibodies that bind only to the target protein.
- Confirms both presence and molecular weight of proteins, helping in accurate identification.
- Very sensitive, capable of detecting even low levels of proteins in a sample.
- Useful for studying protein expression, post-translational modifications, and disease-related proteins.
- Widely applied in clinical diagnosis (e.g., HIV testing), research, and biotechnology.

Disadvantages of Western Blotting

- Time-consuming and multi-step process (gel electrophoresis, transfer, blocking, antibody incubation, detection).
- Requires expensive reagents, antibodies, and specialized equipment.
- Needs high technical skill; small errors can affect results.
- Possibility of non-specific binding and background noise, which may reduce accuracy.
- Limited quantitative accuracy, mainly provides semi-quantitative results.
- Not suitable for high-throughput screening of many samples at once.

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29. Differentiate between C banding and G banding.

Difference between C-banding and G-banding

- **C-banding**
  - Stains constitutive heterochromatin (mainly centromeric regions)
  - Shows dark bands at centromeres and repetitive DNA regions
  - Used to study chromosome structure and centromere abnormalities
- **G-banding**
  - Produces light and dark bands along entire chromosomes
  - Stains AT-rich regions more darkly
  - Used for karyotyping and detecting chromosomal abnormalities (deletions, translocations)

LONG QUESTIONS

1. What is Spectrophotometer?

Spectrophotometer

A spectrophotometer is an analytical instrument used to measure the amount of light absorbed or transmitted by a sample at a specific wavelength. It works on the principle that different molecules absorb light at different wavelengths, and this absorption is used to determine the concentration of substances like DNA, RNA, and proteins in a solution.

Uses

- Measures concentration of biomolecules (DNA, RNA, proteins)
- Checks purity of samples (e.g., A260/A280 ratio for DNA)
- Used in biochemical and molecular biology experiments
- Helps in clinical and research laboratory analysis

2. Write properties of primers seen during primer designing.

Properties of Primers in Primer Designing

- ***Length:*** Primers are usually 18–25 nucleotides long. This length is optimal because it provides good specificity without reducing binding efficiency.
- ***GC content:*** Should be around 40–60%. This ensures stable binding between primer and template DNA due to proper hydrogen bonding.
- ***Melting temperature (Tm):*** Forward and reverse primers should have similar Tm (generally 50–65°C) so they anneal properly during PCR cycles.
- ***Specificity:*** Primers must be specific to the target sequence to avoid non-specific amplification of unwanted DNA regions.
- ***3' end stability:*** The 3' end is crucial for DNA polymerase extension; it should be stable and often ends with G or C for better binding.
- ***Avoid secondary structures:*** Primers should not form hairpins, self-dimers, or cross-dimers, as these reduce efficiency of amplification.
- ***Avoid repeats:*** Long repetitive sequences should be avoided because they can cause mispriming and non-specific amplification.

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3. Explain the process of lysis of RBC in DNA extraction organic method.

Lysis of RBC in Organic DNA Extraction

In the organic method of DNA extraction, RBC (red blood cell) lysis is an initial step used to remove red blood cells and isolate white blood cells (which contain DNA).

Process

- Whole blood is mixed with RBC lysis buffer containing ammonium chloride or Tris-based solution.
- The buffer causes osmotic imbalance, leading to bursting of RBCs.
- RBCs break down because they do not have nuclei, so they are easily lysed.
- After incubation, the mixture is centrifuged.
- The white blood cells (WBCs) form a pellet, while lysed RBC contents remain in the supernatant.
- The supernatant is discarded, and the WBC pellet is used for further DNA extraction steps.

4. Define: Multiplex PCR, Nano assisted PCR, Quantitative PCR.

1. Multiplex PCR

Multiplex PCR is a modified PCR technique in which two or more different DNA targets are amplified simultaneously in a single reaction tube using multiple sets of primers. It is used to detect several genes or pathogens at the same time, saving time, sample, and reagents. It is widely used in diagnosis, genetic testing, and pathogen detection**.**

2. Nano-assisted PCR

Nano-assisted PCR is an advanced form of PCR in which nanoparticles (such as gold, silver, or silica nanoparticles) are added to the reaction mixture to improve PCR performance. These nanoparticles enhance thermal conductivity, enzyme stability, and primer specificity, resulting in faster, more sensitive, and more accurate DNA amplification. It helps reduce non-specific amplification and improves yield.

3. Quantitative PCR (qPCR)

Quantitative PCR (also called real-time PCR) is a technique used to amplify and simultaneously measure the amount of DNA in real time during the PCR process. It uses fluorescent dyes or sequence-specific probes that emit signals proportional to DNA amplification. It is widely used for gene expression analysis, viral load detection, and DNA quantification with high accuracy and sensitivity.

5. What is the purpose of DNA precipitation step and how it is carried out?

Purpose of DNA Precipitation

- DNA precipitation is used to separate and concentrate DNA from the aqueous solution after extraction.
- It helps in removing contaminants such as salts, proteins, phenol, and detergents.
- Converts dissolved DNA into a visible pellet form, making it easy to handle and purify.
- Produces high-quality DNA suitable for PCR, sequencing, and other molecular techniques.

How DNA Precipitation is Carried Out

- After DNA extraction, the sample contains DNA in an aqueous solution.
- Add salt solution (e.g., sodium acetate or NaCl) to neutralize the negative charge of DNA phosphate backbone.
- Slowly add cold ethanol (95–100%) or isopropanol in equal or greater volume.
- Mix gently by inversion (avoid vigorous shaking to prevent DNA shearing).
- Keep the tube at low temperature (ice or −20°C for 10–30 minutes) to enhance precipitation.
- Centrifuge at high speed to collect DNA as a white pellet at the bottom of the tube.
- Carefully discard the supernatant without disturbing the pellet.
- Wash the pellet with 70% ethanol to remove remaining salts and impurities.

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- Centrifuge again and remove wash solution.
- Air-dry the pellet briefly to remove alcohol.
- Finally dissolve DNA in TE buffer or nuclease-free water for storage and further use.

6. What type of agarose gel is used in electrophoresis? Also write procedure for preparation of agarose gel.

Type of Agarose Gel Used in Electrophoresis

- Standard agarose gel (0.7%–2%) is used for DNA electrophoresis.
- Low percentage gel (0.7–1%) → separates large DNA fragments
- High percentage gel (1.5–2%) → separates small DNA fragments
- Usually 1% agarose gel is most commonly used for routine DNA analysis

Procedure for Preparation of Agarose Gel

- Weigh required amount of agarose powder (e.g., 1 g for 1% gel in 100 mL buffer).
- Add agarose to TAE or TBE buffer in a conical flask.
- Heat the mixture in a microwave or hot plate until agarose dissolves completely (clear solution).
- Allow the solution to cool to about 50–60°C.
- Add ethidium bromide or safe DNA stain (if required) and mix gently.
- Pour the solution into a gel casting tray with a comb placed.
- Allow it to solidify at room temperature (20–30 minutes).
- Remove the comb carefully and place gel in electrophoresis tank for use.

7. What are reagents of PCR and how thermocycler is setup?

Reagents of PCR

- ***Template DNA:*** Contains the target sequence to be amplified.
- ***Forward and reverse primers:*** Short oligonucleotides that define the DNA region to be amplified.
- ***Taq DNA polymerase:*** Heat-stable enzyme that synthesizes new DNA strands.
- ***dNTPs (dATP, dTTP, dGTP, dCTP):*** Nucleotide building blocks for DNA synthesis.
- ***PCR buffer:*** Maintains optimal pH and ionic strength for enzyme activity.
- ***$MgCl_2$:*** Essential cofactor that activates DNA polymerase.
- ***Nuclease-free water:*** Used to adjust final reaction volume.

Thermocycler Setup

- ***Program setup:*** Switch on thermocycler, create a new program, and enter reaction volume (20–50 µL) and cycle parameters.
- ***Initial denaturation:*** Set 94–95°C for 2–5 minutes to separate double-stranded DNA completely.
- ***Denaturation step:*** Set 94–95°C for 30 seconds to separate DNA strands in each cycle.
- ***Annealing step:*** Set 50–65°C for 30 seconds to allow primers to bind to target DNA.
- ***Extension step:*** Set 72°C for 30–60 seconds for DNA synthesis by Taq polymerase.
- ***Number of cycles:*** Set 25–35 cycles depending on required amplification.
- ***Final extension:*** Set 72°C for 5–10 minutes to complete unfinished DNA strands.
- ***Hold step:*** Set 4°C to preserve PCR products until retrieval.
- ***Loading:*** Place PCR tubes properly and start the thermocycler run.

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8. Explain process for isolation of RNA using Trizol.

RNA Isolation using TRIzol (Process)

- ***Sample collection:*** Collect fresh cells or tissue and keep them RNase-free to prevent RNA degradation.
- ***Cell lysis:*** Add TRIzol reagent to the sample to break cells and release RNA, DNA, and proteins while inactivating RNases.
- ***Homogenization:*** Mix thoroughly to ensure complete disruption and uniform solution formation.
- ***Phase separation:*** Add chloroform, shake gently, and centrifuge to separate the mixture into three layers: RNA in the upper aqueous phase, DNA in the interphase, and proteins in the lower organic phase.
- ***RNA precipitation:*** Transfer the aqueous phase and add isopropanol to precipitate RNA as a visible pellet.
- ***Centrifugation:*** Spin the sample to collect RNA as a pellet at the bottom of the tube.
- ***Washing:*** Wash the pellet with 70% ethanol to remove salts and impurities.
- ***Drying:*** Air-dry the RNA pellet briefly to remove residual ethanol.
- ***Dissolution:*** Dissolve RNA in RNase-free water or buffer for downstream applications.
- ***Storage:*** Store purified RNA at −20°C or −80°C for long-term use.

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